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Experiment Name: Experiment to understand data preprocessing in Data Science.

```
import numpy as np
import pandas as pd
df = pd.read_csv('pre-process_datasample.csv')
df
df.info()
df.Country.mode()
df.Country.mode()[0]
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(round(df['Salary'].mean()))
df
pd.get_dummies(df.Country)
updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2,
3]]],axis=1)
updated_dataset
updated_dataset['Purchased'] = updated_dataset['Purchased'].replace( ['No',
'Yes'], [0, 1])
updated_dataset
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     str
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 452.0 bytes
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0

	France	Germany	Spain	Age	Salary	Purchased
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	False	True	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1